

Total No. of Questions : 8]

SEAT No. :

P-6665

[Total No. of Pages : 5

[6181]-231

B.E. (Mechanical)

**HEATING, VENTILATION, AIR CONDITIONING AND
REFRIGERATION**

(2019 Pattern) (Semester - VII) (402041)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Use of scientific calculator is allowed.
- 5) Assume Suitable data, if necessary.

- Q1)** a) Explain with a neat sketch Thermostatic Expansion Valve. [6]
b) Explain with a neat sketch Low Pressure (LP) cut off used in VCR cycle. [6]
c) Distinguish between the air cooled and water cooled condensers. [5]

OR

- Q2)** a) Explain with neat sketch flooded type evaporator. [6]
b) Explain with neat schematic the frost control circuit used in VCR cycle. [6]
c) Explain with neat schematic CO₂ transcritical cycle. [5]

- Q3)** a) Explain load calculations factors for air conditioning. [8]
b) Atmospheric air at 30°C dry bulb temperature and 75% relative humidity, enters a cooling coil at a rate of 200 m³/min. The coil dew point temperature is 14°C and the by-pass factor of the coil is 0.1. Determine : [10]
i) The temperature of air leaving the cooling coil;
ii) The capacity of the cooling coil in TR and in kW;
iii) The amount of water vapour removed per minute; and
iv) The sensible heat factor for the process.

P.T.O.

OR

Q4) a) Explain the terms : [9]

- i) GRSHF
- ii) BPF
- iii) ERSHF

b) A conference room for sitting 100 persons is to be maintained at 22°C DBT and 60% relative humidity. The outdoor conditions are 40°C DBT and 27°C WBT. The various loads in the auditorium are as follows : [9]

- i) Sensible and latent heat loads per person 80 W and 50 W respectively;
- ii) Lights and fans, 15000 W;
- iii) Sensible heat gain through glass ceiling etc. 15000W.
- iv) The air infiltration is 20 m³/min and fresh air supply is 100 m³/min.
- v) Two-third of recirculated room air and one-third of fresh air are mixed before entering the cooling coil.
- vi) The bypass factor of the coil is 0.1.

Determine :

- a) Apparatus dew point,
- b) Grand total heat load and
- c) Effective room sensible heat factor

Q5) a) What is infiltration and Ventilation? What are different Methods of Infiltration? [8]

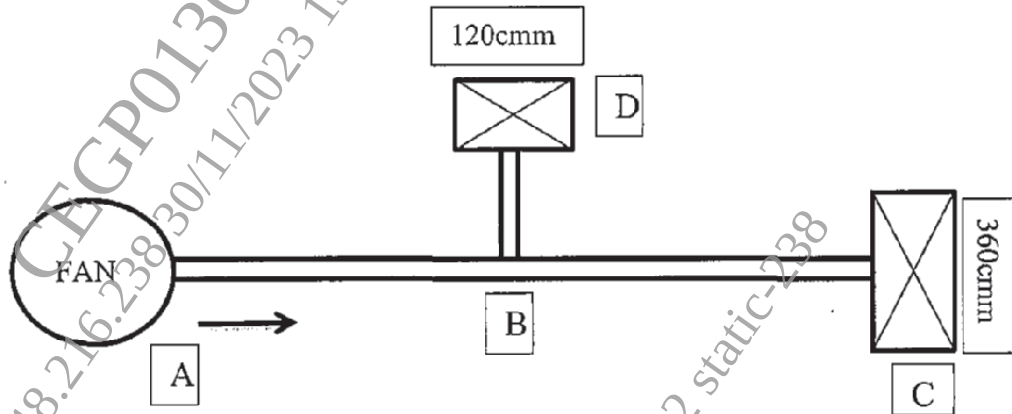
b) A circular duct of $\phi 400$ mm is selected to carry air at a velocity 440 m/min. If duct is replaced by rectangular duct of aspect ratio 1.5. Find the size of rectangular duct for equal friction when: [10]

- i) Velocity in two Duct is same.
- ii) Discharge in two Duct is same.

If $f = 0.015$, Find the pressure loss per 100m length of Duct. Take density of air = 1.15 kg/m³.

OR

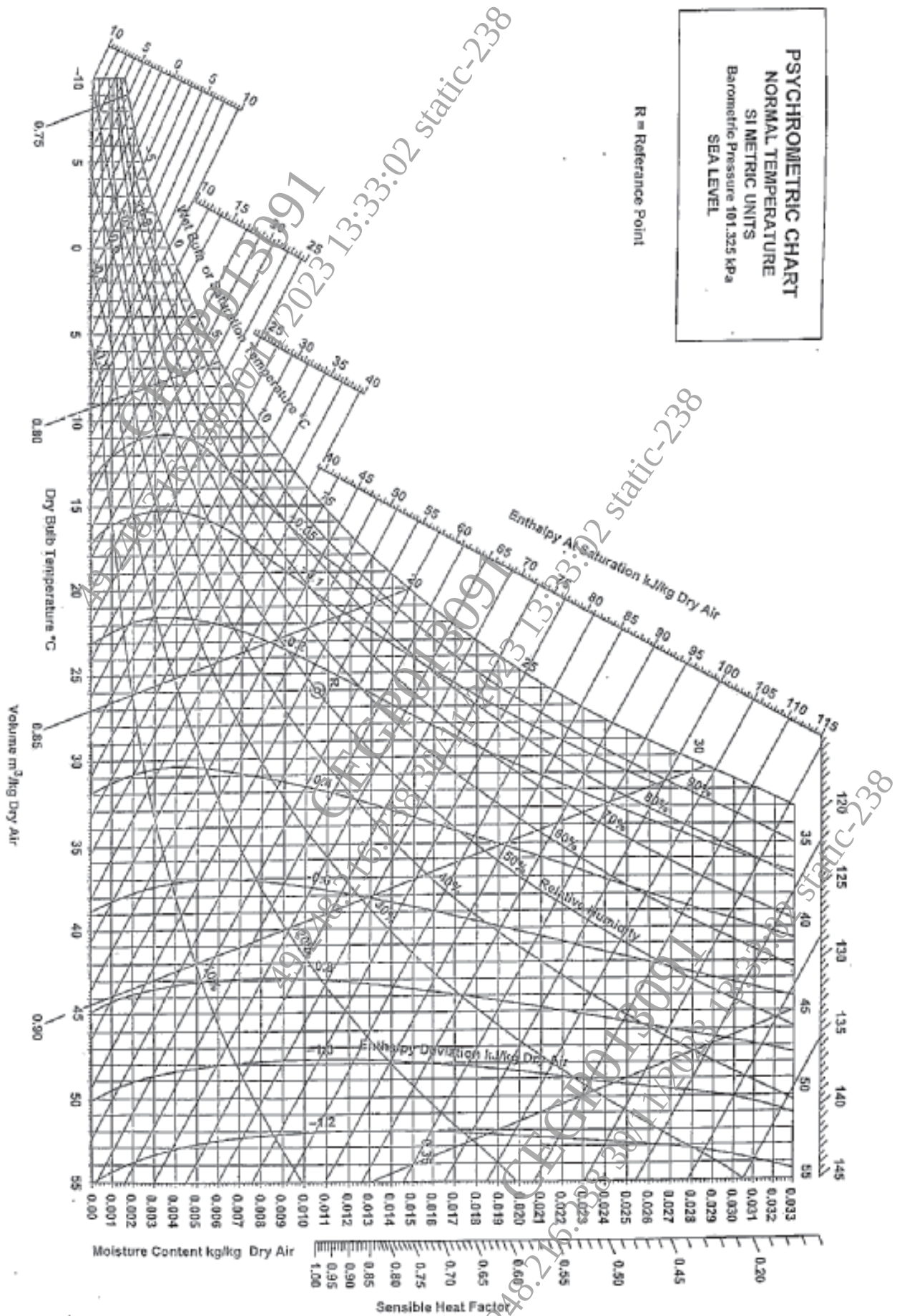
- Q6) a)** Explain Natural Ventilation and Mechanical Ventilation. [8]
- b)** Using Equal friction method, determine the duct diameter and velocity for section AB, BD and BC. Assume velocity in the main duct AB = 600 m/min. Also Calculate maximum pressure drop in the duct system. Distance AB = 30 m, Distance BC = 30 m and Distance BD = 10 m. Refer the figure as given below. [10]



- Q7) a)** Explain with neat sketch winter Air conditioning system. [6]
- b)** Draw and Explain water to water heat pump circuit. [6]
- c)** Write a short note on solid packed tower. [5]

OR

- Q8) a)** Explain with neat sketch All water system. [6]
- b)** Write a short note on Sorbents and Desiccants. [6]
- c)** Write a short note on Radiant cooling. [5]



Friction Chart for Circular Ducts

